ECO735 Replication Assignment

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1.

TABLE I: 1998 AVERAGE PUPIL AND SCHOOL CHARACTERISTICS, PRE-TREATMENT

	Group 1 (25 schools)	Group 2 (25 schools)	Group 3 (25 schools)	Group 1− Group 3	Group 2− Group 3	Group 1− Group 2
Panel A: Preschool to Grade 8						
Male	0.534	0.501	0.517	0.016	-0.017	0.033***
				(0.123)	(0.127)	(0.002)
Proportion girls <13 years and all boys	0.741	0.718	0.723	0.018*	-0.005	0.022**
				(0.059)	(0.633)	(0.018)
Grade Progression	-1.749	-1.695	-1.776	0.028	0.081**	-0.054*
				(0.387)	(0.013)	(0.096)
Year of birth	1983.756	1983.847	1983.827	-0.071	0.019	-0.090
				(0.329)	(0.787)	(0.263)
Panel B: Grades 3–8						
Attendance recorded in school registers	0.973	0.963	0.969	0.004***	-0.006***	0.010***
				(0.016)	(0.000)	(0.000)
Access to latrine at home	0.822	0.808	0.818	0.004	-0.010	0.014*
				(0.616)	(0.243)	(0.091)
Have livestock at home	0.659	0.673	0.664	-0.005	0.008	-0.014

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	Group 1 (25 schools)	Group 2 (25 schools)	Group 3 (25 schools)	Group 1– Group 3	Group 2– Group 3	Group 1– Group 2
				(0.612)	(0.408)	(0.177)
Weight-for-age Z-score	-1.390	-1.402	-1.440	0.050***	0.038**	0.012
				(0.007)	(0.042)	(0.505)
Blood in stool	0.264	0.221	0.193	0.070***	0.027***	0.043***
				(0.000)	(0.002)	(0.000)
Sick often	0.097	0.104	0.081	0.016***	-0.023***	-0.007
				(0.007)	(0.000)	(0.269)
Malaria/ fever in the last week	0.368	0.384	0.402	-0.034***	-0.018*	-0.016
				(0.001)	(0.093)	(0.110)
Clean	0.603	0.657	0.670	-0.067***	-0.012	-0.054***
				(0.000)	(0.224)	(0.000)
Panel C: School characteristics						
District exam score, 1996; grades 5-8	-0.197	0.175	0.021	-0.218	0.154	-0.372
				(0.312)	(0.537)	(0.130)
Distance to Lake Victoria	10.034	9.923	9.455	0.578	0.467	0.111
				(0.732)	(0.824)	(0.956)
Pupil population	392.720	403.800	375.880	16.840	27.920	-11.080
				(0.704)	(0.661)	(0.863)
School latrines per pupil	0.007	0.006	0.007	0.001	0.000	0.001
				(0.393)	(0.697)	(0.238)
Proportion moderate-heavy infections in zone	0.0369	0.367	0.360	0.009	0.006	0.002
				(0.771)	(0.827)	(0.935)
Group 1 pupil within 3 km	461.059	408.290	344.508	116.552	63.782	52.769
				(0.334)	(0.578)	(0.680)

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	Group 1 (25 schools)	Group 2 (25 schools)	Group 3 (25 schools)	Group 1− Group 3	Group 2− Group 3	Group 1− Group 2
Group 1 pupil within 3-6 km	844.515	652.033	869.654	-25.139	-217.622*	192.483
				(0.872)	(0.088)	(0.178)

Notes: p-values in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

(d)

Pre-treatment balance across the three groups appears largely satisfactory. Most baseline characteristics show no statistically significant differences between groups at 5% or 1% significance levels. However, some variables do exhibit marginal or statistically significant differences, particularly at the 10% level. These differences are not systematic across variables and are consistent with what one would expect due to random sampling variation. There is no consistent pattern indicating that one group is systematically advantaged or disadvantaged across multiple variables.

2.

TABLE VIII: SCHOOL PARTICIPATION, SCHOOL-LEVEL DATA

	Group 1 (25 schools)	Group 2 (25 schools)	Group 3 (25 schools)		
Panel A: First year post-treatment	*1st-year treatment*	*Comparison*	*Comparison*	*Group1-(Groups 2&3)*	*Group2-Group 3*
Girls <13 years, and all boys	0.848	0.743	0.789	0.084**	-0.047
				(0.035)	(0.038)
Girls >=13 years	0.913	0.845	0.876	0.053**	-0.031
				(0.026)	(0.043)
Preschool, Gr 1, Gr 2 in early 1998	0.781	0.687	0.712	0.082	-0.025
				(0.052)	(0.046)
Gr 3, Gr 4, Gr 5 in early 1998	0.898	0.807	0.866	0.063**	-0.059*
				(0.026)	(0.031)
Gr 6, Gr 7, Gr 8 in early 1998	0.955	0.883	0.925	0.051***	-0.042
				(0.018)	(0.028)
Recorded as 'dropped out' in early 1998	0.020	0.023	0.027	-0.005	-0.005
				(0.009)	(0.013)
Females	0.854	0.760	0.801	0.076**	-0.041
				(0.033)	(0.041)
Males	0.855	0.751	0.810	0.076**	-0.059
				(0.034)	(0.039)
Panel B: Second year post-treatment	*2nd year Treatment*	*1st year treatment*	*Comparison*	*Group1-Group 3*	*Group2-Group 3*
Girls <13 years, and all boys	0.734	0.722	0.672	0.062**	0.049
				(0.030)	(0.030)
Girls >=14 years	0.683	0.689	0.629	0.054*	0.060*

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	Group 1 (25 schools)	Group 2 (25 schools)	Group 3 (25 schools)		
				(0.030)	(0.035)
Preschool, Gr 1, Gr 2 in early 1998	0.720	0.733	0.656	0.064*	0.077**
				(0.033)	(0.031)
Gr 3, Gr 4, Gr 5 in early 1998	0.768	0.777	0.754	0.036	0.044*
				(0.024)	(0.026)
Gr 6, Gr 7, Gr 8 in early 1998	0.788	0.775	0.754	0.034	0.021
				(0.028)	(0.033)
Recorded as 'dropped out' in early 1998	0.168	0.109	0.052	0.116**	0.057**
				(0.046)	(0.027)
Females	0.718	0.721	0.631	0.087***	0.090**
				(0.032)	(0.036)
Males	0.723	0.702	0.665	0.058*	0.037
				(0.030)	(0.031)

SE values in parentheses

(a)

The equation used is:

$$Y_{ijt} = \alpha + \beta_1 T_{1it} + \beta_2 T_{2it} + X'_{ijt}\delta + \sum_d (\gamma_d \cdot N^T_{dit}) + \sum_d (\phi_d \cdot N^T_{dit}) + u_i + e_{ijt}$$

Here, Y is the individual education outcome. Here, i refers to the school, j refers to the student and $t = 1, 2$ denotes the period of treatment. T_{1it} and T_{2it} are indicator variables for assignment of school for the first and second years. N^T_{dit} is the number of pupils within treatment school T inside a certain distance d. N_{dit} is the total primary school pupils with the same distance d. γ_d measures the cross-school de-worming externalities at distance d. X_{ijt} is a vector of characteristics of the pupil while u_i accounts for correlation within the same school. e_{ijt} is, as usual, the error term for an individual.

(b)

- For girls below 13 and all boys, the difference in school participation for the five post-treatment participation observations in the first year after medical treatment is 8.4 percentage, and is significant at 95% confidence interval. The same difference between Group 1 and 3 in 1999 is 6.2% and that between Group 2 and 3 is 4.9%. This difference is less among older girls in Panel A at 5.3 percentage points. For lower grades in Panel A, the differences (8.2 percentage points and 6.3 percentage points) are greater than for higher grades (5.1 percentage points). Hence, roughly 5-10 percentage points increase is observed

in school participation due to treatment of de-worming. Group 1 also consistently has higher magnitude of school participation rates than Group 3 in both Panels. Average participation rates do fall from 1998 to 1999.

- Most of the differences are statistically significant at 95% or 99% confidence level. Hence, we cannot reject the null hypothesis that the treatment has no effect on school participation.

- It is possible that the elasticity of school participation with respect to health is higher for pupils in younger grades. There are positive externalities due to this treatment. The people have more benefit than cost for the de-worming treatment, which is probably why the takeup is large.

- There are larger effects for younger pupil and boys and younger girls. There is a positive effect for older girls but there may be some externalities because they were not directly treated. This can be due to reduced transmission of diseases due to others taking the treatment.